Count Data Models

Miguel García Moreno

15 February 2026

Abstract

This project studies a count data set with excess zeros and fits models based on count-data techniques. After a data-quality check and an exploratory analysis, Zero-Altered (ZA / hurdle) and Zero-Inflated (ZI) models are estimated. Among the candidates, the best model is selected using AIC and BIC.

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1 Exploratory analysis

First, data quality is checked for missing values. The data set contains no missing observations. After that check, an exploratory analysis is carried out.

Figure 1 shows a clear excess of zero observations in this count sample. Because of that feature, ZA and ZI models are used to model the data appropriately.

The same figure also shows that a theoretical negative binomial distribution adapts well to the empirical distribution of the counts. The negative binomial parameters are obtained by maximum likelihood on the positive sample values (zeros excluded). Poisson appears to fit worse: its theoretical shape does not match the empirical distribution as closely. Overdispersion in the sample is an additional reason to prefer the negative binomial.

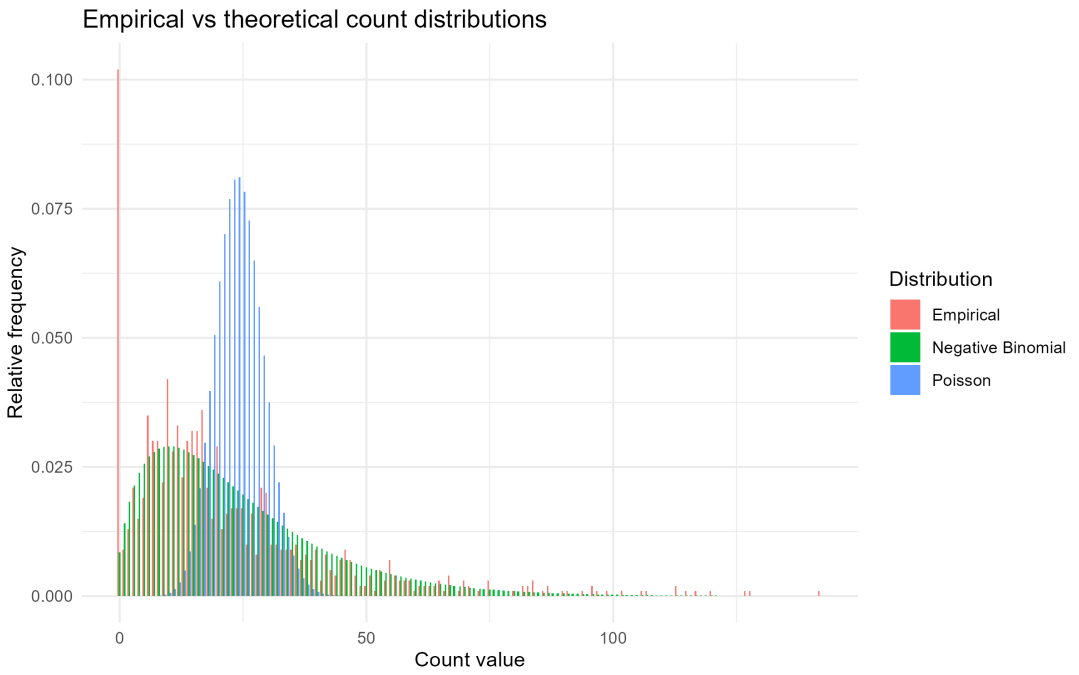


Figure 1: Empirical relative frequencies compared with theoretical Poisson and negative binomial distributions.

2 Model fitting

ZA and ZI models are fitted starting from saturated specifications and removing variables with high individual p-values (non-significant coefficients). After searching over count and zero-process distributions, the shortlisted ZA models are:

`zanbp_x12f3_x1235f2`, `zanbb_x12f3_x5`, `zapp_x123f3_x1235f2`, `zapb_x123f3_x5`.

Naming convention: `ZA` denotes a Zero-Altered (hurdle) model; the next letter(s) give the count distribution (`nb` = negative binomial, `p` = Poisson); the following letter is the zero-process distribution; after the first underscore come the count-model covariates, and after the second underscore the zero-model covariates. Thus `zanbp_x12f3_x1235f2` is a ZA model with negative binomial counts, Poisson zeros, covariates `x1`, `x2`, `factor3` in the count part, and `x1`, `x2`, `x3`, `x5`, `factor2` in the zero part.

The shortlisted ZI models are:

`zinb_x12f3_0f1`, `zip_x132f3_0f1`, `zinb_x12f3_1`.

For ZI models only the count distribution is varied, given their theoretical construction.

Goodness-of-fit comparisons show a substantial gap between ZA and ZI models (Table 1): ZA models have markedly lower AIC/BIC and recover a large share of the observed zeros, whereas the ZI candidates predict none. This supports interpreting the sample zeros as arising from a single process (structural / "real" zeros) rather than a separate inflation process.

Table 1: Performance of shortlisted ZA and ZI models on the full sample, ordered by ascending AIC. Residual SD is the standard deviation of model residuals; predicted zeros count fitted values rounded to zero.

Model	Residual SD	AIC	BIC	Obs. zeros	Pred. zeros
`zanbb_x12f3_x5`	5.78	5432.7	5467.1	102	66
`zanbp_x12f3_x1235f2`	6.02	5453.1	5507.1	102	46
`zapb_x123f3_x5`	5.56	5453.5	5487.8	102	66
`zapp_x123f3_x1235f2`	5.80	5473.9	5527.9	102	46
`zinb_x12f3_1`	10.48	6035.0	6064.4	102	0
`zinb_x12f3_0f1`	10.48	6038.8	6078.1	102	0
`zip_x132f3_0f1`	10.42	6059.5	6098.7	102	0

3 Model selection

AIC and BIC are used for selection (avoiding reliance on the Vuong test). By that criterion (Table 1) the best model is `zanbb_x12f3_x5`, with the lowest AIC and BIC among the candidates. It predicts 66 of the 102 zeros present in the sample. Figure 2 shows its diagnostic plots.

Predictions lie close to the observed counts, with more dispersion at larger count values: low counts are predicted more accurately than high ones. Points concentrate around the perfect-prediction line $y = x$. The residual standard deviation is approximately 5.78, which is relatively low given the range of the response. Overall, the model is a good fit for this data set.

The exploratory analysis already suggested that the negative binomial would match the empirical distribution best; the selected model indeed uses a negative binomial count component.



Figure 2: Observed vs fitted and observed vs residuals for the selected model `zanbb_x12f3_x5`.

4 Appendix: Code

The full analysis script is `AEM_02_Count_regression.R` in this folder. Running it (from the folder or with the path set accordingly) loads `AEM_02_Count_regression_and_GAM.xlsx`, fits the ZA/ZI models, writes the comparison table under `tables/`, and regenerates the figures under `figures/` used in this report.